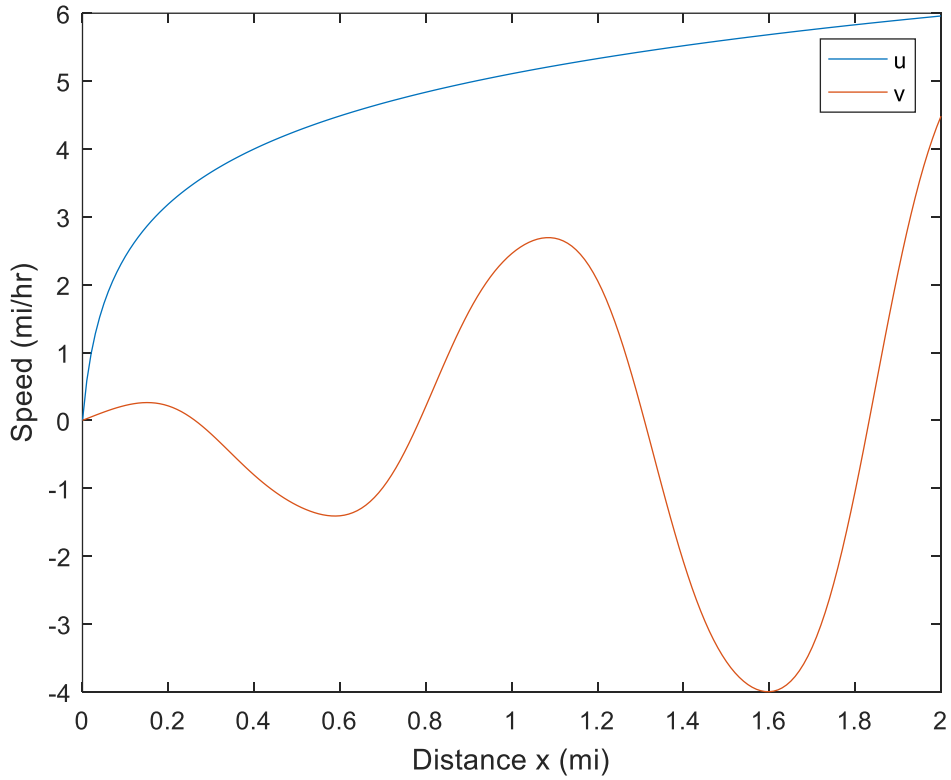
1. Use MATLAB to plot the function $u = 2\log_5(60x + 1)$ and $v = 3x \cdot \sin(\cos(6x))$ over the interval $0 \le x \le 2$. Properly label the plot and each curve. The variables u and v represent speed in miles per hour; the variable x represents distance in miles. (25%)



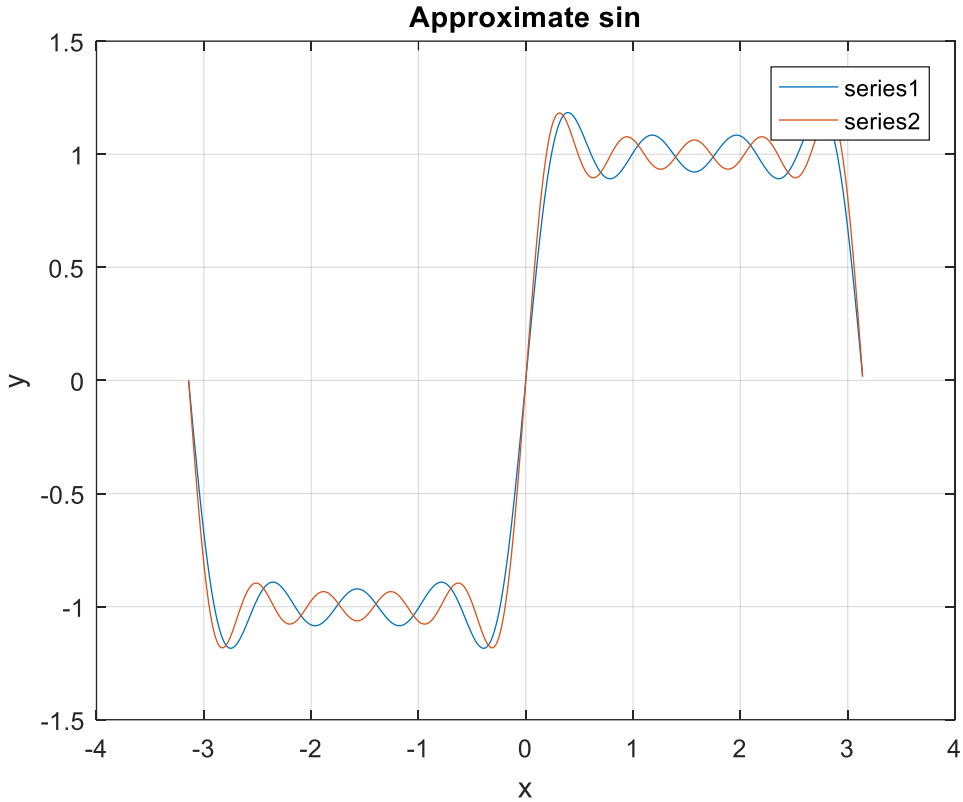
2. The Fourier series is a series representation of a periodic function in terms of sines and cosines. The Fourier series representation of the function

$$f(x) = \begin{cases} 1 & 0 < x < \pi \\ -1 & -\pi < x < 0 \end{cases}$$

is

$$\frac{4}{\pi}\left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \cdots\right)$$

Plot on the same graph its two series representations, one using the previous **four** terms, the other using the previous **five** terms. (25%)



3. Write a MATLAB assignment statement for each of the following functions, assuming that *w*, *x*, *y*, and *z* are row vectors of equal length and that *c* and *d* are scalars.

   Assuming that the elements of *w* lie between 0 and 1 with increment 0.1 (in total, 11 points); the elements of *x* lie between 0.5 and 0.6 with increment 0.01; the elements of *y* lie between 1 and 11 with increment 1; the elements of *z* lie between 10 and 110 with increment 10; *c* = 2.8; *d* = 3.5. (25%)

$$f = \frac{1}{\sqrt{2\pi(c/x)}}$$

$$E = \frac{(x+w)/(y+z)}{x+w/(y-z)}$$

$$A = \frac{e^{-d/(2x)}}{(\ln y)\sqrt{cz}}$$

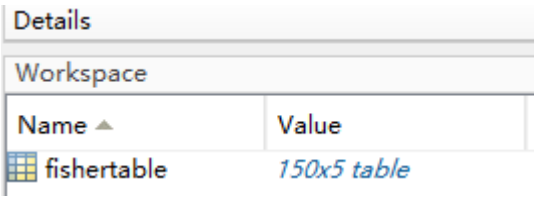
$$S = \frac{x(2.15+0.35y)^{1.8}}{y(1-x)^{z}}$$

```
f =

    0.1686    0.1703    0.1719    0.1736    0.1752    0.1768    0.1784    0.1800    0.1816    0.1831    0.1847


E =

    0.0909    0.0550    0.0426    0.0362    0.0322    0.0294    0.0274    0.0258    0.0244    0.0234    0.0224


A =

       Inf    0.0062    0.0034    0.0025    0.0021    0.0018    0.0016    0.0015    0.0014    0.0013    0.0013


S =

   1.0e+43 *

    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0001    8.1439
```

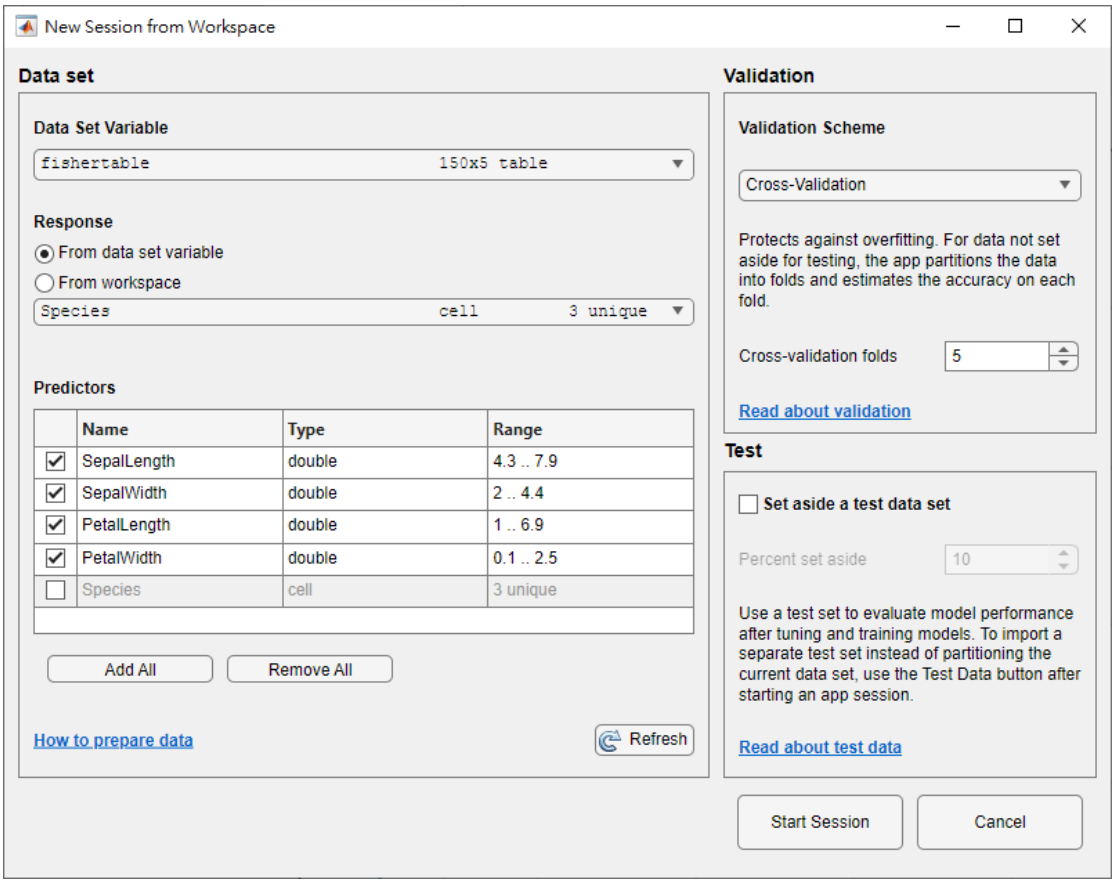
4. Use the Classification Learner App to train Classification Models. The database is fisheriris.csv.

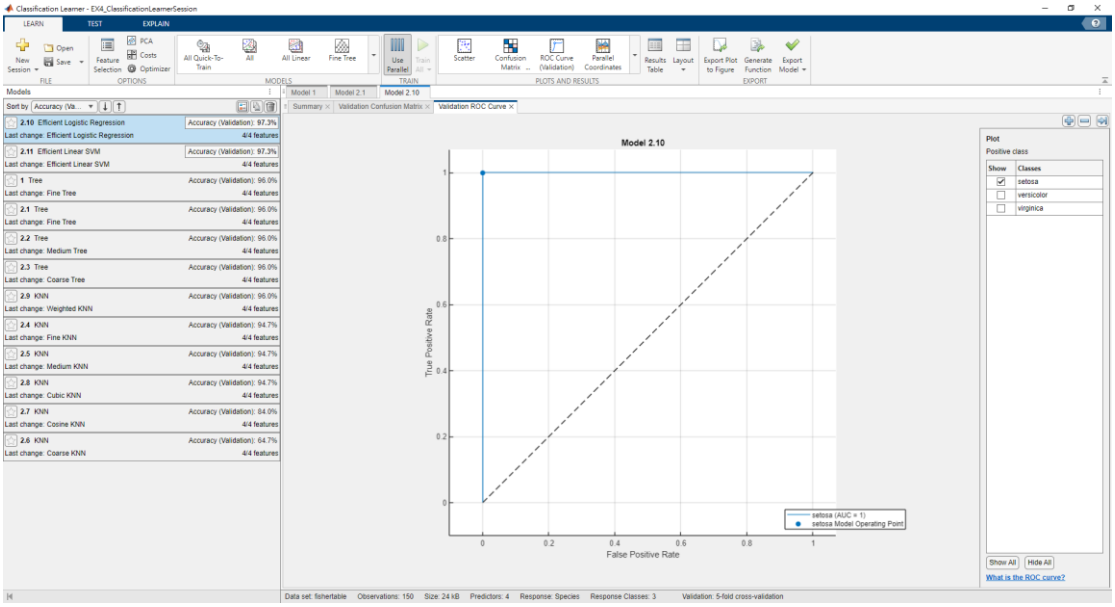
   At the command window, write the statement: fishertable = readtable('fisheriris.csv');

   Then, in your workspace, you have a variable: fishertable

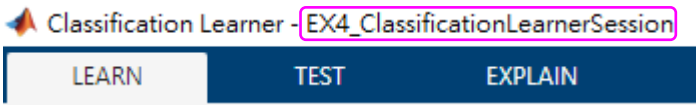


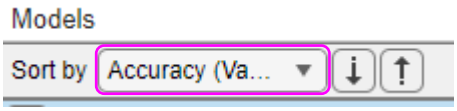
   Output:

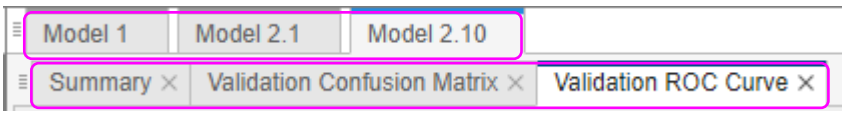




Some details to be created and shown:



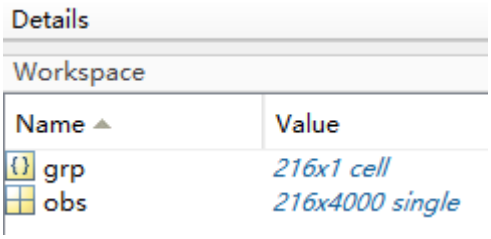




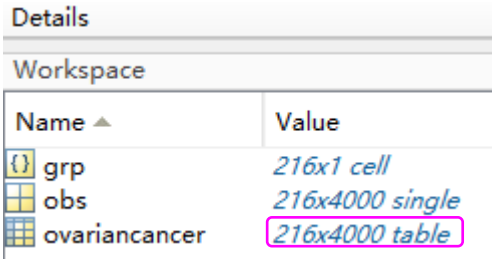
5. Use the Classification Learner App to train Classification Models. The database is ovariancancer.mat.

   At the command window, write the statement: load ovariancancer

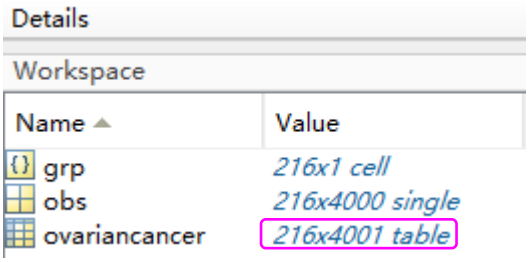
   Then, in your workspace, you have two variables: grp and obs



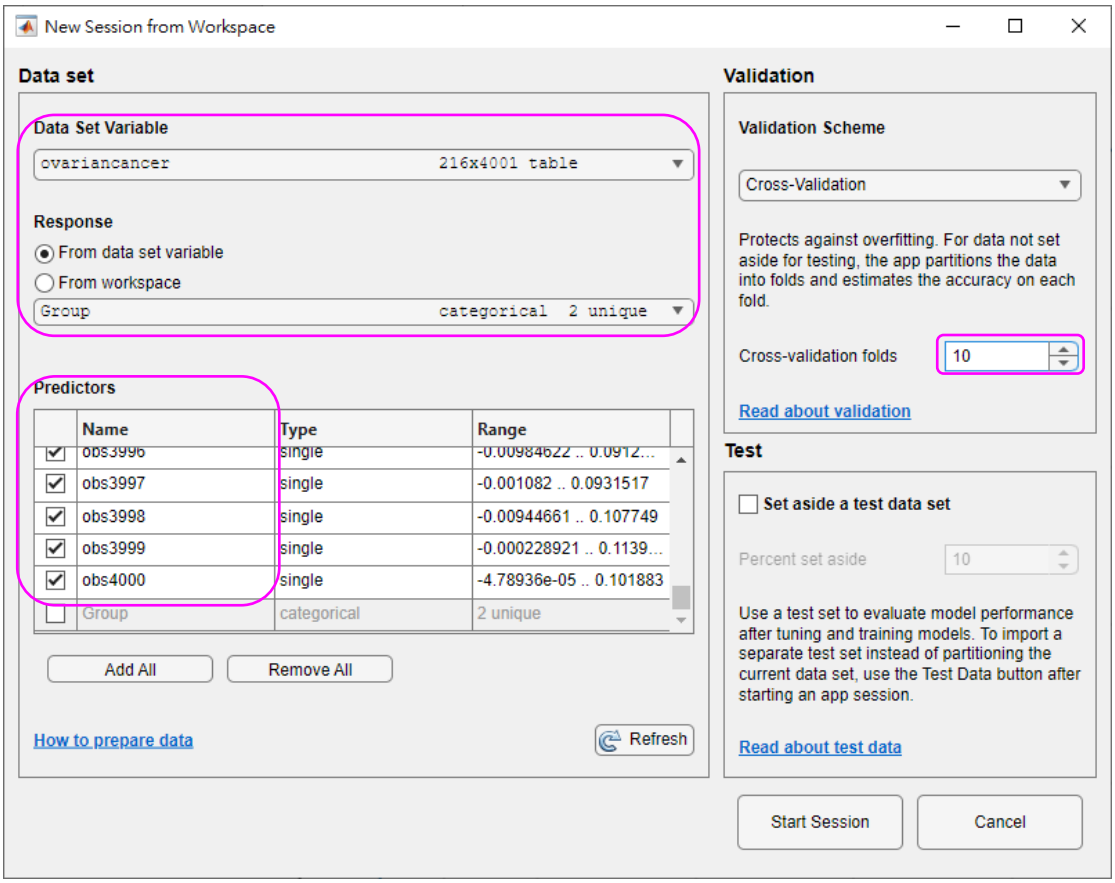
   Use a built-in function to transfer an array (obs) to a table, and then you have three variables: grp, obs, and ovariancancer (216 × 4000)

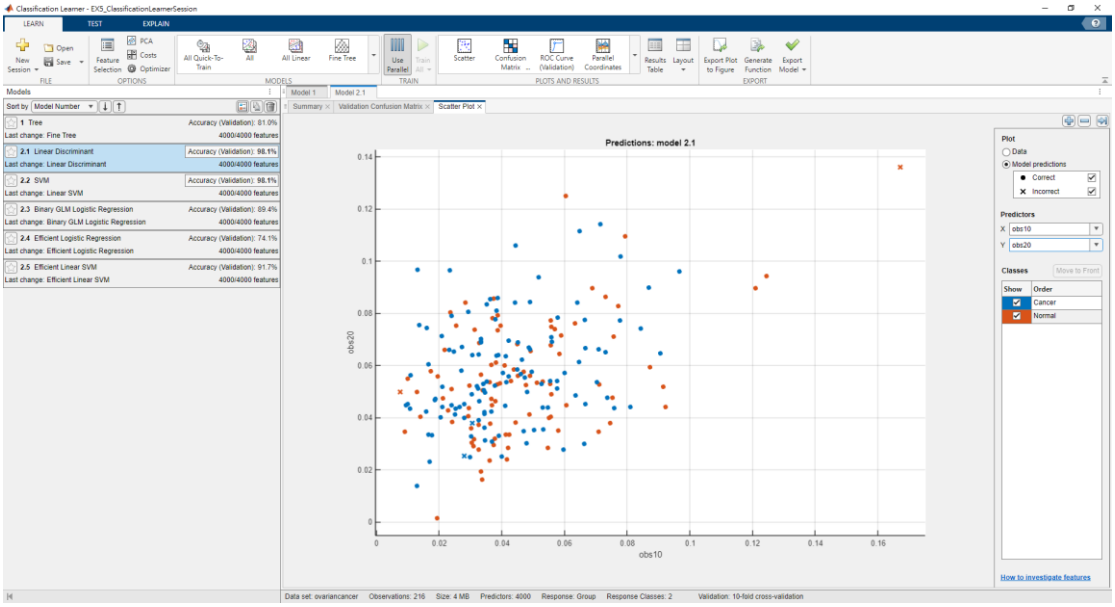


   Add an extra column called Group to ovariancancer; the column is equal to categorical(grp); and then you have changed the size of the variable ovariancancer (216 × 4001)

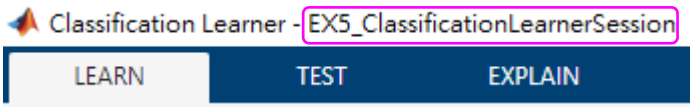


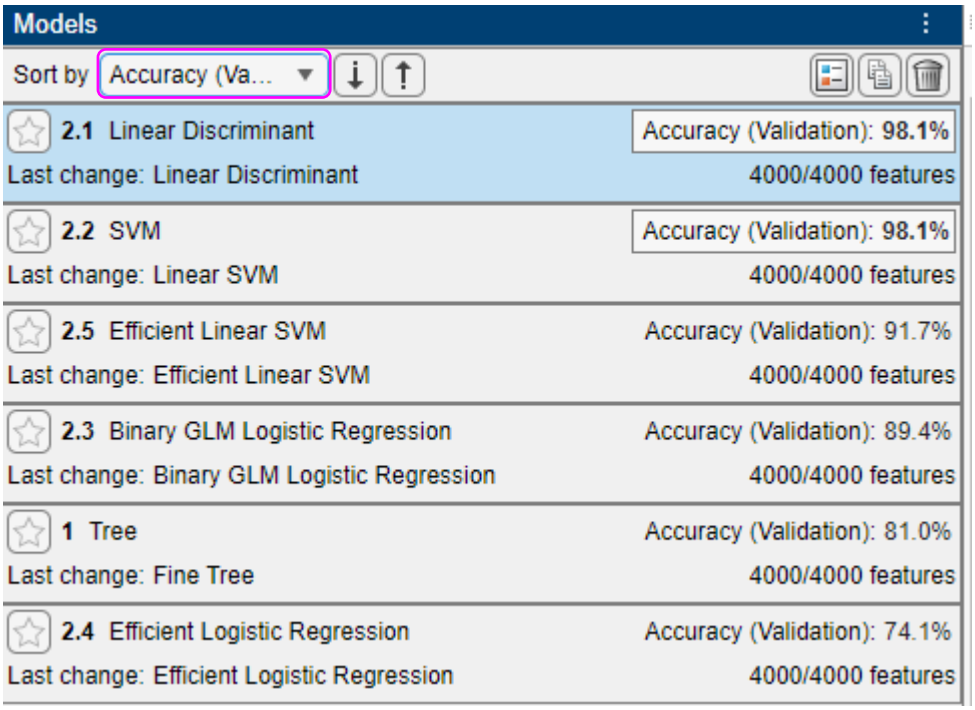
   Output:

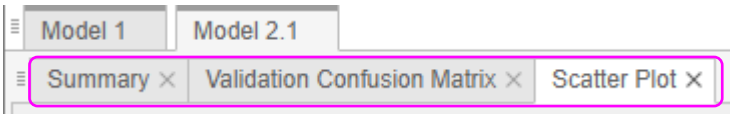


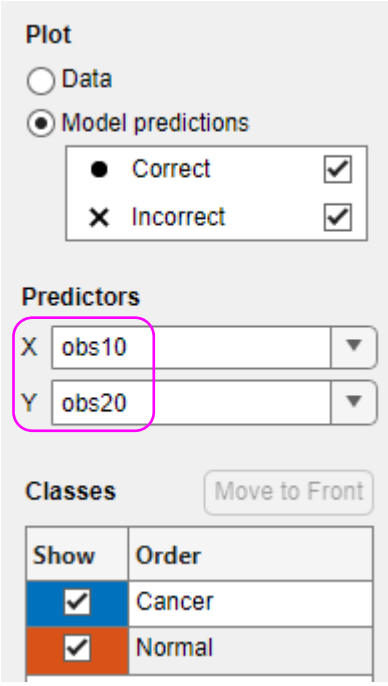


Some details to be created and shown:




Models
Sort by
Accuracy (Va...
2.1 Linear Discriminant
Accuracy (Validation): 98.1%
Last change: Linear Discriminant
4000/4000 features
2.2 SVM
Accuracy (Validation): 98.1%
Last change: Linear SVM
4000/4000 features
2.5 Efficient Linear SVM
Accuracy (Validation): 91.7%
Last change: Efficient Linear SVM
4000/4000 features
2.3 Binary GLM Logistic Regression
Accuracy (Validation): 89.4%
Last change: Binary GLM Logistic Regression
4000/4000 features
1 Tree
Accuracy (Validation): 81.0%
Last change: Fine Tree
4000/4000 features
2.4 Efficient Logistic Regression
Accuracy (Validation): 74.1%
Last change: Efficient Logistic Regression
4000/4000 features


Model 1
Model 2.1
Summary
Validation Confusion Matrix
Scatter Plot


Plot
Data
Model predictions
Correct
Incorrect
Predictors
X
obs10
Y
obs20
Classes
Move to Front
Show
Order
Cancer
Normal